

**M A S A R Y K
U N I V E R S I T Y**

FACULTY OF INFORMATICS

**Musikk: A Modern, Social-First
Music Streaming Platform**

Bachelor's Thesis

KIRILL VOROZHTSOV

Brno, Fall 2025

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Advisor: Mgr. Luděk Bártek, Ph.D

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Declaration

Hereby I declare that this paper is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

During the preparation of this thesis, the following AI tools were used:

- **ChatGPT:** Used for debugging and making small code corrections, as well as LaTeX formatting.
- **Grammarly:** Used for text correction.

Kirill Vorozhtsov

Advisor: Mgr. Luděk Bártek, Ph.D

Acknowledgements

These are the acknowledgements for my thesis, which can span multiple paragraphs.

Abstract

This bachelor's thesis aims to design and implement a web-based music streaming platform focused on social interactions between users.

The theoretical part presents a survey that analyzes people's music listening habits and shows that such a platform could find its user base. It then compares existing music-related services with built-in social features and derives functional and non-functional requirements for the proposed system. Based on these requirements, suitable technologies and architectural patterns are selected.

The practical part covers the technical part of the platform, for example, data model, audio processing, user interaction features, real-time synchronization, etc.

The result is a fully functional and deployed application.

Keywords

Audio Streaming, Python, JavaScript, MPEG-DASH, HLS, WebSockets

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1 Introduction

Music streaming has been significantly growing in popularity each year for the past decade.[2] And this is to be expected; numerous studies showed how important music is for people. For instance, Rentfrow in his 2012 paper shows that it can influence mood, change self-perception, and help with social bonding. In addition, the rise of the clubbing scene, music festivals, and other social music-related activities clearly indicates the important social role of music.[4]

Despite this, it is surprising that features that facilitate social interactions are not widely implemented in the existing music streaming platforms, as will be shown in ??.

The goal of this thesis is to design a music-centric platform that provides capabilities for collaboration and social interaction around music. This work is divided into the following six chapters:

1. 2 Presents the outcomes of a survey illustrating how people consume music, how prevalent it is in social interactions, and why this type of platform can be relevant.
2. ?? Explores what existing streaming platforms and other music-related services already offer.
3. ?? Outlines the functional and non-functional criteria for the application.
4. ?? Describes the choices of technologies that are used by the application.
5. ?? Explains the development process and implementation details.
6. ?? Summarizes the results of the work.

2 Music Consumption Survey

In order to better support the choice of the topic and show that a music streaming platform centered around social interactions could be relevant as a service, a brief survey was conducted. It examined individual music listening and discovery habits, platform usage patterns, music-related people interactions and the attitude towards possible social features on the platform.

2.1 Methods

The service chosen for the questionnaire was ‘Google Forms’[1], as it provides a simple interface for survey creation, allows for easy sharing of the form, and support result extraction as a ‘csv’ file.

The questionnaire itself consisted of 15 questions. Most of them are multiple-choice and closed-ended, with some allowing a custom answer. Custom answers are grouped under the “other” answer in the provided figures. Answers in their original form can be found in the full survey.

The complete questionnaire is provided in Appendix B; an online version is also available.¹

2.2 Results

A total of 119 people participated, with the majority being from Russian-speaking countries. The majority were in the 18 to 30 age group, with the exact distribution shown in Figure 2.1.

This section presents the survey outcomes and analyzes their significance. Not everything is discussed; for example, questions dedicated to music discovery are omitted.

1. https://docs.google.com/forms/d/1vhhAu_SfuHV4xy6JoDaRsM5uCElYn_csTmN8u4ZlpHc

How old are you?

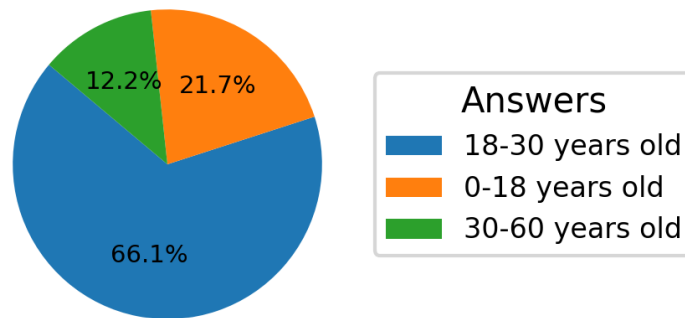


Figure 2.1: Age distribution of respondents

2.2.1 Engagement

Firstly, it was necessary to determine whether people actually do listen to music frequently and whether they use streaming platforms to do so.

Over 50% of respondents reported listening to more than 500 songs per month (Figure 2.2), and nearly 80% stated that they listen to music daily (Figure 2.3).

The following logical step was to determine how the audio content is mostly accessed.

2.2.2 Listening Methods

As can be seen in Figure 2.4, the percentage of streaming platform usage across all ages is nearing 100%, with slightly higher numbers in the lower age groups. Although downloaded files and physical media have their presence, especially for people aged 30-60, it is usually only an auxiliary option next to streaming solutions.

The question dedicated to the popularity of individual platforms has shown that Spotify is the most used one, which aligns with global statistics[5]. However, Yandex Music, being in second place, deserves an explanation. As mentioned previously, most of the respondents are from Russian-speaking countries, specifically Russia. With many

2. MUSIC CONSUMPTION SURVEY

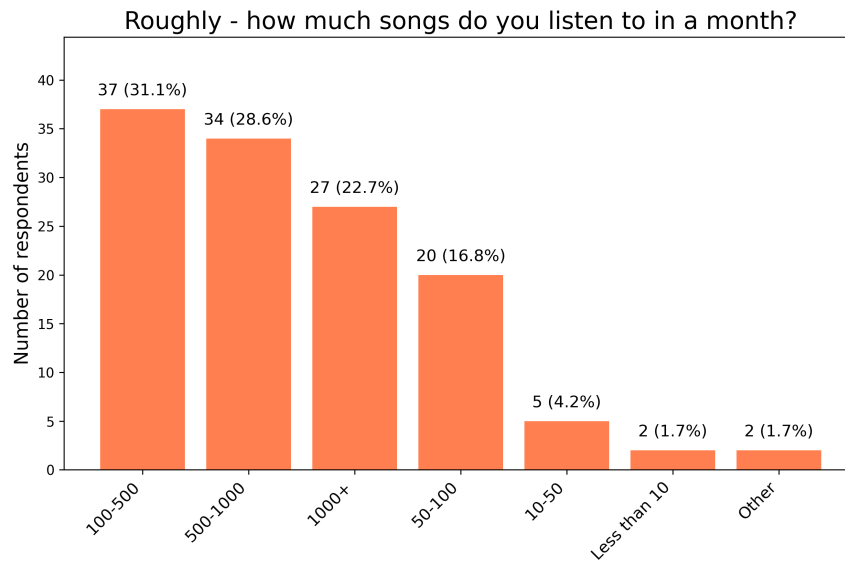


Figure 2.2: Monthly song count per respondent

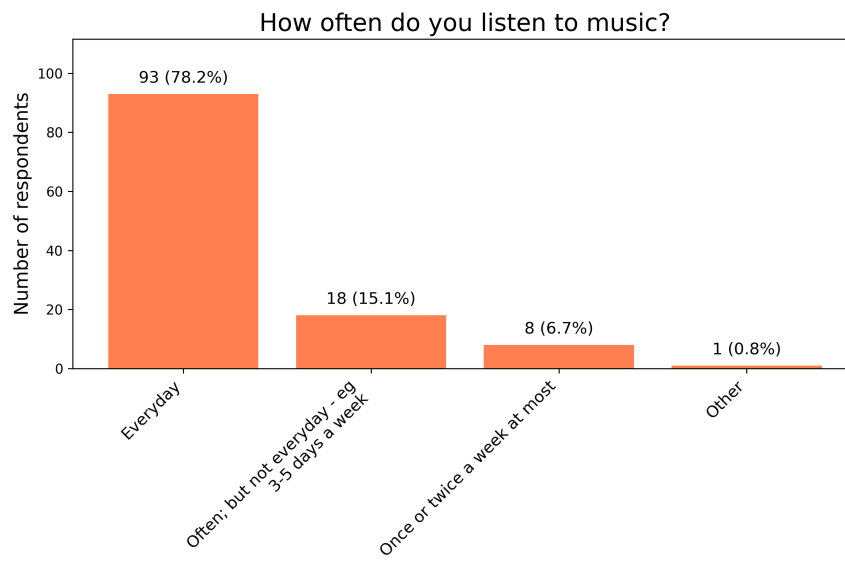


Figure 2.3: Listening frequency of respondents

2. MUSIC CONSUMPTION SURVEY

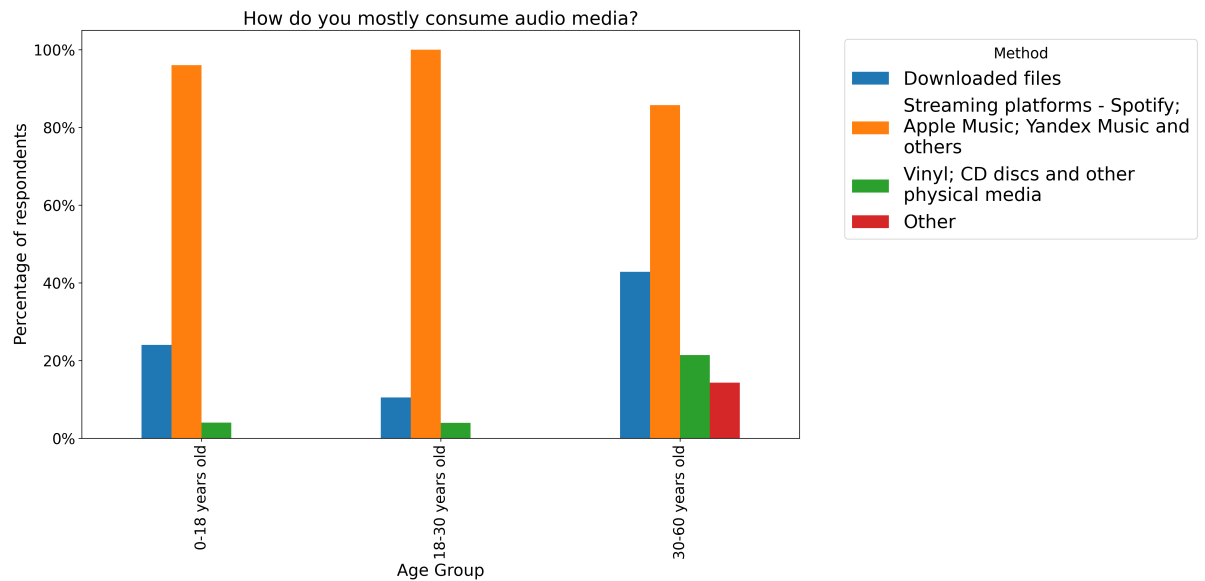


Figure 2.4: Audio media consumption by age group

Western companies leaving its market in 2022, music streaming services included, most users have moved to locally available products: Yandex Music, VK Music, Zvuk, and others. Another notable point is the absence of other popular region-specific platforms, such as Amazon Music for the US market, QQ Music for the Chinese market and JioSaavn for the Indian one, as all of the respondents were based in the European part of the world. The complete statistics are shown in Figure 2.5.

The survey showed that streaming services are indeed widely used. The next important step was to analyze the regularity of music-centered social interactions.

2.2.3 Social Interactions

The results of Question 10 indicate that nearly all respondents (99.2%) reported listening to music with others, primarily in offline settings such as gatherings or car rides (Figure 2.6). Online collaborative listening methods, while less common, were also mentioned by a quarter of respondents, showing that digital solutions for shared listening are used, although not as prevalently as in-person scenarios.

2. MUSIC CONSUMPTION SURVEY

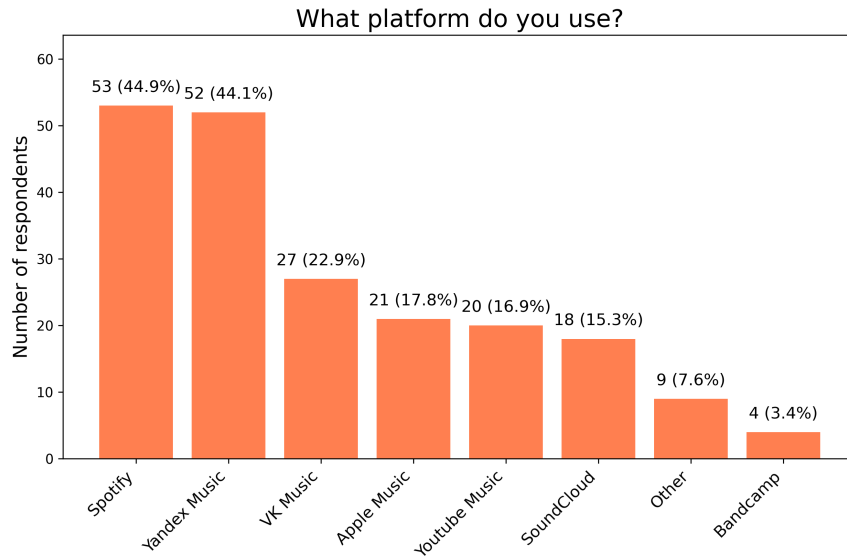


Figure 2.5: Streaming platforms popularity

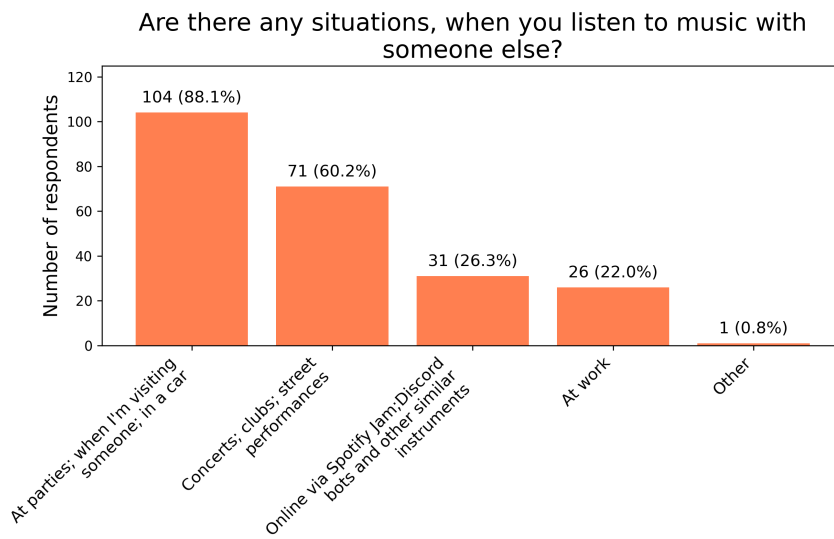


Figure 2.6: Social interaction methods

Question 11 (Figure 2.7) investigates how often these interactions occur: results show that a significant portion of respondents engage in shared listening regularly. Around 60% reported doing so at least a few times per month, with a smaller group indicating almost daily interactions. This supports the previously mentioned idea (1) that music consumption is a common social activity.

How often do you listen to music with someone else?



Figure 2.7: Social interactions frequency

The next question is phrased as follows: “How often do you/your friends share/discuss music?”. While the previous question focused on real-time in-person collective listening, this one highlights more intentional and in-depth musical interactions, such as sharing songs, recommending music, or discussing it. Music, in that case, is not just the background but the main purpose of engagement. The same trend as before can be observed in the responses: over 70% (Figure 2.8) indicated that they share music at least several times a month, with many doing so weekly or more often. This further emphasizes the active social role of music.

Question 13 — “How does that happen?” (as in music sharing) — provides an insight into the specific means of sharing. Many respondents (Figure 2.9) stated that they typically send links from streaming platforms through external messaging apps. While this shows an apparent demand for sharing music online, it also reveals a gap: these interactions are fragmented across platforms. Hence, enabling such exchanges directly within the streaming app could provide a smoother and a more uniform experience.

Responses to the question “Would you say that you know a lot of people with similar music taste?” were mixed, with a near-even split. This

2. MUSIC CONSUMPTION SURVEY

How often do you/your friends share/discuss music?

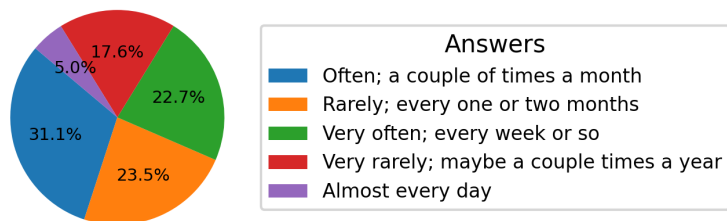


Figure 2.8: Music sharing frequency

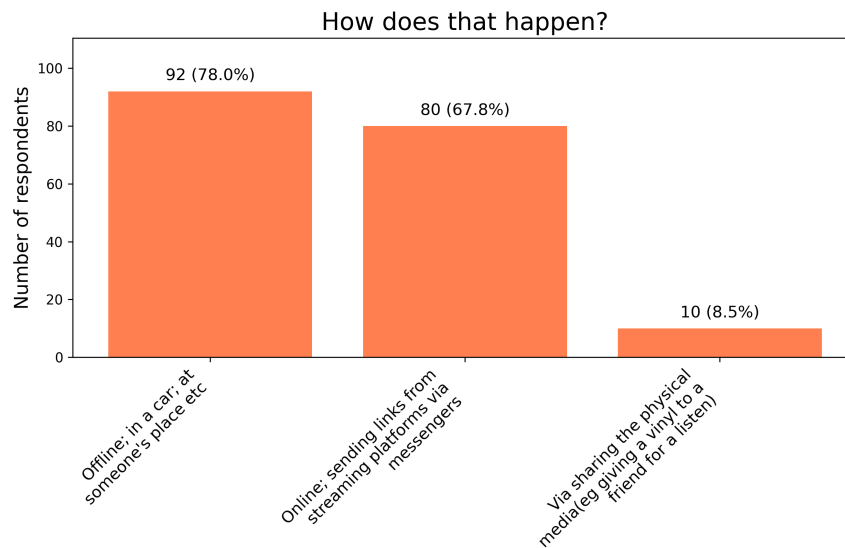


Figure 2.9: Music sharing methods

suggests that while some users already have a social circle with similar music taste, many do not. Interestingly, when asked *“Would you like to connect with others who share your musical taste, or influence those around you to explore and appreciate the music you enjoy?”*, the majority answered positively. This indicates an interest in the expansion of musical connections and supports the idea that social discovery features could be useful within a streaming platform. This is further supported by the responses to Question 15, where over 60% of participants stated they would use additional social features if available on a streaming platform.

2.3 Summary

Overall, the results show that people listen to music frequently and that streaming services are the main medium to do so. It is also evident that music consumption is frequent, and music often appears in social interactions. Yet, current platforms only partly support interactions with other users. Therefore, the idea of a music streaming service with integrated social features could be relevant, addressing real user needs.

A An appendix

B Survey Questionnaire

Music Consumption Study

This short form is designed to analyze the people's behaviour related to music - ways of discovering new audio pieces, the role of music in social interactions, preferred ways of consuming audio information etc.

1. How old are you?

Mark only one oval.

- ☐ 0-18 years old
☐ 18-30 years old
☐ 30-60 years old
☐ 60+ years old

2. How do you mostly consume audio media?

Tick all that apply.

- ☐ Streaming platforms - Spotify, Apple Music, Yandex Music and others.
☐ Vinyl, CD discs and other physical media.
☐ Downloaded files.
☐ Other: _____

3. How often do you listen to music?

Tick all that apply.

- ☐ Everyday.
☐ Often, but not everyday - e.g. 3-5 days a week.
☐ Once or twice a week at most.
☐ Only on special occasions.
☐ Other: _____

B. SURVEY QUESTIONNAIRE

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

4. Roughly - how much songs do you listen to in a month?

Tick all that apply.

- ☐ 1000+
☐ 500-1000
☐ 100-500
☐ 50-100
☐ 10-50
☐ Less than 10
☐ Other: _____

5. Would you say that you are consistent with the music you listen to? I.e. do you mostly listen to the same artists/songs/genres?

Mark only one oval.

- ☐ Yes
☐ Mostly
☐ Rather no than yes
☐ No

6. How often do you listen to something new?

Mark only one oval.

- ☐ Very rarely, once a couple of months.
☐ Rarely, maybe a handful times a month.
☐ Pretty often, every week or so.
☐ Often, a couple of times a week.
☐ Every or almost every day.

B. SURVEY QUESTIONNAIRE

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

7. How do you find out about new music?

Tick all that apply.

- ☐ At random - in cafes, shops and other venues.
- ☐ From people I know.
- ☐ Non-specialized online platforms - Instagram, TikTok, Youtube(videos), etc.
- ☐ Specialized online platforms - Rate Your Music, Pitchfork, Wire, etc.
- ☐ Games, movies, other forms of digital content.
- ☐ Buying new physical releases(vinyl, cds ...)
- ☐ Recommendations on streaming platforms
- ☐ Other: _____

8. Would you say that you know a lot of people with similar music taste?

Mark only one oval.

- ☐ Yes
- ☐ No

9. Would you like to connect with others who share your musical taste, or influence those around you to explore and appreciate the music you enjoy?

Mark only one oval.

- ☐ Yes
- ☐ No, I already have enough people around me with whom I can share the music I like
- ☐ No, I do not feel the need in that
- ☐ Other: _____

B. SURVEY QUESTIONNAIRE

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

10. Are there any situations, when you listen to music with someone else?

Tick all that apply.

- ☐ At parties; when I'm visiting someone; in a car.
- ☐ Online via Spotify Jam, Discord bots and other similar instruments.
- ☐ Concerts, clubs, street performances.
- ☐ At work.
- ☐ Other: _____

11. How often do you listen to music with someone else?

Mark only one oval.

- ☐ Very rarely, maybe a couple times a year.
- ☐ Rarely, every one or two months.
- ☐ Often, a couple of times a month.
- ☐ Very often, every week or so.
- ☐ Almost every day.
- ☐ Other: _____

12. How often do you/your friends share/discuss music?

Mark only one oval.

- ☐ Very rarely, maybe a couple times a year.
- ☐ Rarely, every one or two months.
- ☐ Often, a couple of times a month.
- ☐ Very often, every week or so.
- ☐ Almost every day.
- ☐ Other: _____

B. SURVEY QUESTIONNAIRE

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

13. How does that happen?

Tick all that apply.

- ☐ Online, sending links from streaming platforms via messengers.
- ☐ Offline, in a car, at someone's place etc.
- ☐ Via sharing the physical media(e.g. giving a vinyl to a friend for a listen)
- ☐ Other: _____

14. Question for users of streaming platforms.

What platform do you use?

Tick all that apply.

- ☐ Spotify
- ☐ Apple Music
- ☐ Yandex Music
- ☐ VK Music
- ☐ SoundCloud
- ☐ Bandcamp
- ☐ Youtube Music
- ☐ Other: _____

15. Question for users of streaming platforms.

Would you benefit, if current streaming platforms added extra social features(i.e. would you use them)? For example, chat discussions for playlists and songs, feeds with music added by the people you follow, forums, direct messages, etc.

Mark only one oval.

- ☐ I would use these features often.
- ☐ I would use these features rarely.
- ☐ I wouldn't use these features.

B. SURVEY QUESTIONNAIRE

Music Consumption Study

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16. Question for users of streaming platforms.

Are there any other features you would like to see?

This content is neither created nor endorsed by Google.

Google Forms

B. SURVEY QUESTIONNAIRE

Music Consumption Study

https://docs.google.com/forms/u/5/d/1vhhAu_SfuHV4xy6JoDaRsM5...

Bibliography

- [1] *Google Forms*. URL: <https://docs.google.com/forms/u/0/> (visited on 04/05/2025).
- [2] *Music Streaming Market Decade Insights*. URL: <https://www.businessofapps.com/data/music-streaming-market/> (visited on 11/17/2025).
- [3] Peter J. Rentfrow. *The Role of Music in Everyday Life: Current Directions in the Social Psychology of Music*. 2012. URL: <https://compass.onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-9004.2012.00434.x> (visited on 03/15/2025).
- [4] *Sharing Music: Social and Communal Aspects of Concert-Going*. URL: <https://ojs.meccsa.org.uk/index.php/netknow/article/view/428/250> (visited on 11/17/2025).
- [5] *Spotify Popularity*. URL: https://simplebeen.com/music-streaming-statistics/?utm_source=chatgpt.com (visited on 04/15/2025).